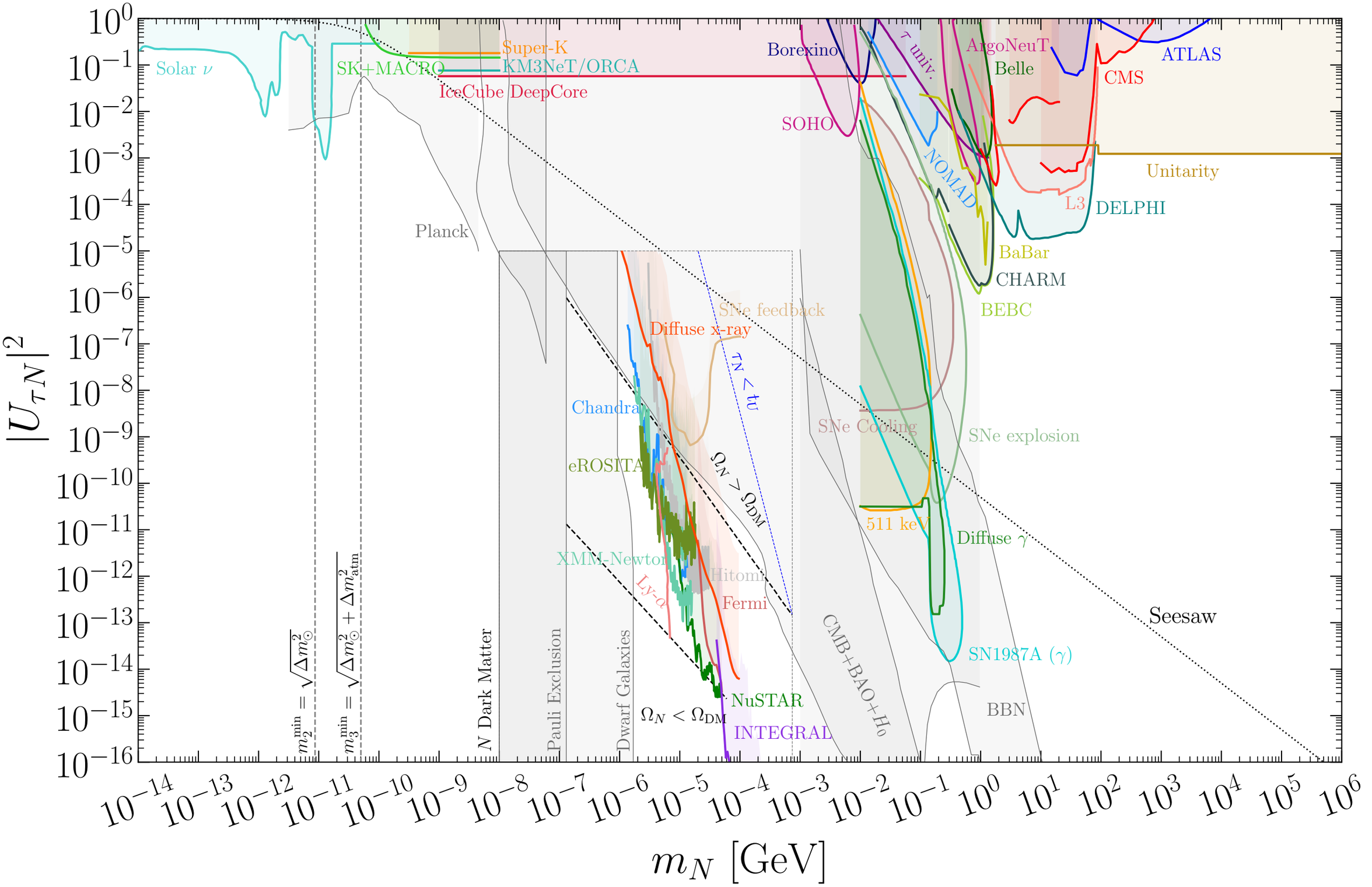

$|U_{\tau N}|^2$
m_N [GeV]
Solar ν
Super-K
SK+MACRO
KM3NeT/ORCA
IceCube DeepCore
Borexino
τ univ.
ArgoNeuT
Belle
ATLAS
CMS
SOHO
NOMAD
Unitarity
L3
DELPHI
BaBar
CHARM
BEBC
Planck
SNe feedback
Diffuse x-ray
$\tau_N < t_U$
Chandra
eROSITA
SNe Cooling
SNe explosion
$\Omega_N > \Omega_{\rm DM}$
511 keV
Diffuse γ
XMM-Newton
Hitomi
Ly-α
Fermi
Seesaw
SN1987A (γ)
$m_2^{\rm min} = \sqrt{\Delta m^2_\odot}$
$m_3^{\rm min} = \sqrt{\Delta m^2_\odot + \Delta m^2_{\rm atm}}$
N Dark Matter
Pauli Exclusion
Dwarf Galaxies
CMB+BAO+H_0
BBN
NuSTAR
$\Omega_N < \Omega_{\rm DM}$
INTEGRAL